

AI Assisted Programming Platform for Education and Assessment



ABSTRACT

Programming education at the university level demands constant feedback and efficient assessment that are difficult to scale. This platform integrates an AI-powered mentor into a browser-based coding environment where students write, run, and submit code across multiple programming languages while receiving real-time Socratic guidance that nudges them toward solutions without revealing answers. Teachers manage assignments, monitor student progress, and use AI assistance for grading, with an exam mode that disables AI features to preserve academic integrity.

System Architecture & Methodology

The platform was developed and deployed as a fully containerized web application, ensuring scalability and a seamless end-to-end experience.

- **Backend:** High-performance API handling code execution and LLM integration.
- **Frontend:** Browser-based coding environment with real-time feedback mechanisms.
- **Security:** Role-based access controls and strict environment isolation for exam sessions.

Key Features

- **Socratic AI Mentor:** Provides guided hints and nudges students toward solutions without revealing the direct answers.
- **Comprehensive Teacher Tools:** Enables automated rubric generation, AI-assisted grading, and real-time class analytics.
- **Academic Integrity (Exam Mode):** Features a dedicated exam mode that disables AI assistance to preserve academic honesty.
- **Multi-Language Environment:** Supports writing, running, and submitting code across five different programming languages.

RESULT & CONCLUSION

Testing confirmed that the Socratic hint system and exam mode restrictions function as intended. The platform successfully bridges the gap between large class sizes and personalized programming instruction by reducing manual grading workloads and providing immediate feedback to students.



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